

AI-Powered Nutrition Agent

IBM watsonx.ai + LangFlow
AICTE Internship | Edunet Foundation

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1. Problem Statement

The Challenge

Good nutrition is the foundation of health, but individuals — especially in India — struggle with diet planning, portion control, and tracking nutritional intake. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without personalized, real-time guidance, many people fail to maintain balanced diets or make informed food choices. The situation is worsened by the diversity of Indian regional cuisines, languages, and cultural dietary restrictions that generic apps completely ignore.

Scale of the Problem

- Over **101 million diabetics** in India require strict diet management
- **40% of Indian children** under 5 are stunted due to malnutrition
- **135 million obese adults** in India lack access to personalized dietary guidance
- Only **1 registered dietitian per 1.27 million people** in India

2. Proposed Solution

The proposed solution is a **multi-agent AI-powered Nutrition Agent** built using LangFlow, IBM watsonx.ai, and IBM Granite 3.2 8B Instruct model. The system delivers personalized, real-time dietary guidance tailored to Indian food culture, regional preferences, and individual health profiles.

Multi-Agent Architecture

- **Nutrition Knowledge Agent:** Retrieves and summarizes nutritional data via RAG from USDA and WHO datasets
- **Diet Recommendation Agent:** Generates personalized weekly meal plans based on age, health conditions, allergies, and fitness goals
- **Health Advisory Agent:** Provides disease-specific dietary guidance (diabetes-friendly, heart-healthy, PCOS, thyroid diets)
- **Food Log & Feedback Agent:** Analyzes logged meals and delivers instant calorie counts and nutritional feedback

3. Technology Stack

Technology	Version / Model	Role in Project
IBM watsonx.ai	Enterprise Platform	Model deployment, embeddings, AI governance
IBM Granite	granite-3-2-8b-instruct	Core LLM for reasoning, summarization, personalization
LangFlow	v1.3+	Visual multi-agent workflow orchestration
RAG Pipeline	FAISS / Chroma	Real-time nutritional data retrieval from knowledge base
IBM Cloud	us-south region	Scalable, secure deployment infrastructure

4. Key Features

- ✓ Personalized Indian diet plans (vegetarian, vegan, Jain, Halal-friendly)
- ✓ BMI calculator with health category interpretation
- ✓ Calorie analysis for 500+ Indian and international foods
- ✓ Family profile management with multi-member diet recommendations
- ✓ Disease-specific diets (diabetes, thyroid, PCOS, heart health, kidney disease)
- ✓ RAG-based accuracy: real nutritional data from USDA/WHO, not hallucinations
- ✓ Customizable AGENT_INSTRUCTIONS for tone, diet specialization, and safety rules

5. Social Impact & Future Scope

This solution directly addresses nutrition inequality in India by providing free, AI-powered dietary guidance to individuals who cannot afford a registered dietitian. The system is especially valuable for

rural communities where healthcare access is limited.

Future Enhancements:

- Integration with smartwatches and fitness trackers for real-time activity-based dietary adjustments
- Multilingual voice assistant supporting Hindi, Tamil, Telugu, Bengali and 7 other Indian languages
- Image-based meal logging — photograph your thali for instant nutritional analysis
- Integration with hospital EHR systems for doctor-approved diet plans
- Regional grocery price integration for budget-conscious meal planning

GitHub Repository:
github.com/AMANRAJ-Droid/nutrition-agent

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